

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 233

THE QUINN PLANET CHECKER

straight lines, circles, mathematical repetition
the half-degree walker



orbital and jungle tiers — every robot a checker

720 TRACKS · 48 DAYS · THE GK3 INSTALL

NATURE DOESN'T DO STRAIGHT

Survey — v1.0 in force

GP-IM-233

Revision v1.0 — 24 August 2026

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VETTED BATCH 32 — PASS · FALSE-POSITIVE TESTING

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 233

PHASE 233: THE QUINN PLANET CHECKER — THE SYMMETRY DRONE (STRAIGHT LINES, CIRCLES, MATHEMATICAL REPETITION; THE HALF-DEGREE WALKER; ORBITAL AND JUNGLE TIERS) — STATUS: CURRENT — PASS / RESOLUTION, CLOUD COVERAGE AND FALSE-POSITIVE TESTING. It started as a way to find aliens and ended as a way to find anything. Instead of billion-dollar lidar satellites: a cheap, short-lived survey drone searching for three things in symmetry — straight lines, circles, and mathematically spaced repetition. Nature abhors the straight line; intelligence loves it. The orbital tier walks a half-degree at a time — 720 tracks, about 48 days of coverage, roughly 55 km per step at the equator — reading continent-scale geometry; the tactical tier (quad and GK3) reads the small objects. The doctrinal gem seats as law: thermal finds heat, and heat hides — canopy, rain, insulation, dawn; geometry finds structure, and structure cannot hide. The honest corrections are already in the file: you cannot see a 10 cm patch from orbit because the algorithm knows rectangles; nature has mimics — dunes, basalt polygons, ripples — so the discriminator requires straightness plus scale mix plus context. And the GK3 install runs the identifiers in reverse: every robot a checker, and the expected straight line becomes the crack detector — deviation is damage.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-233, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the checker law as seated (contents line, the doctrine — the three targets, the doctrinal gem, the two tiers, the new-world law — the GK3 install rulings) — §2 the physics, graded — §3 the system in the sky — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 233: **The Planet Checker — The Symmetry Drone (Straight Lines,**

Circles, Mathematical Repetition; the Half-Degree Walker; Orbital and Jungle Tiers)

THE CORE OBJECTIVE (verbatim): 'Core Objective: a cheap, short-lived survey drone — orbital tier for continents, tactical tier for jungle — that searches any surface for the three signatures of intelligence and industry: straight lines, circles, and evenly or mathematically spaced repetition. Check before contact; scan before you settle; find the artifact, not the heat.'

Survey Baseline (graded this sitting): the half-degree walker is sound mechanics and prettier than specified — 720 tracks for full coverage, ~48 days at standard low-orbit period, and 0.5 degrees at the equator is ~55 km, so a cheap 55+ km wide-swath camera achieves CONTIGUOUS single-pass coverage on the same walk. The free upgrade: don't burn propellant to alter course — tune the period and inclination so the orbit walks itself (Landsat lineage: 233 orbits / 16-day repeat, same arithmetic). The detection suite is sixty-year-old tech: Hough transform for lines and circles, FFT frequency peaks for regular spacing. The algorithm is the product; the satellite is a bus; the lifespan is one mapping pass.

Live Instruction Confirmed: Yes — 3 August 2026, morning sitting.

Live instruction, 3 August 2026.

"ok, pre existing or existing intelligent life drones. sure we could spend billions trying to send or carry lidar satellites or we can invent the planet checker drone, a satellite that simply alters course orbit by a half a degree each rotation,"

"not complex, its is searching for 3 things in symmetry, straight lines circles and evenly or mathematically spaced repetition on the surface, cheap because its lifespan only needs to be for the mapping check. if i am in the jungle i dont want thermal goggles, i want these, because it will find everything from your rifle to your silly arm patch"

THE DOCTRINE — AS GIVEN, STRUCTURE READ (KIMI) (VERBATIM)

- **The three targets — AFFIRMED; nature abhors the straight line.** Straight: roads, runways, canals, fence lines, cut trails. Circles: center-pivot irrigation, storage tanks, roundabouts, cook pots. Even spacing: orchards, solar farms, wind farms, city grids, streetlight rows, tent lines, patrol formations. This is exactly how space archaeology finds lost cities from orbit — regular geometry in organic chaos — and how searchers find crashed aircraft: the straight edge that doesn't belong. The technosignature literature says the same for worlds: geometry is the first thing to scan for.
- **The doctrinal gem — SEATED AS LAW.** Thermal finds HEAT, and heat hides: canopy, rain, insulation, dawn. Geometry finds ARTIFACT, and SHAPE IS THE ONE SIGNATURE FUNCTION CAN'T SURRENDER — a road that curves like a river stops being a road; a rifle bent like a vine stops being a rifle. Camouflage defeats color and heat; it cannot defeat straightness without breaking the tool. 'If I am in the jungle I don't want thermal goggles, I want these' — graded correct for the daylight fight;

thermal keeps the night. Complements, not rivals.

- **Two tiers, one algorithm — the honest resolution grade.** ORBITAL TIER: cities, roads, fields, solar farms, camps, cut lines — continent-scale, cheap bus, one pass. TACTICAL TIER: a cheap quad at 100 metres with the same software — the rifle barrel as a sub-pixel straight edge, the arm patch as a rectangle in leaf-chaos, the picket line as regular spacing. 'From your rifle to your silly arm patch': NO from orbit (resolution physics — a patch is 10 cm, cheap orbital is metres), YES from the drone. Detection, not recognition: the algorithm flags 'artifact here'; a human names it.
- **The planet checker at a new world — the Highway doctrine completed.** First thing dropped at any new system body, before contact, before outposts, before landings: the checker. Forty-eight days later the map says whether anyone is home, ever was home, or the ground is yours to read — and the same map feeds the siting law (232: map before you build) and the landing doctrine (227: lights before landings). Scan, then map, then light, then land — the arrival stack.

PHYSICS NOTE — THE AUDIT (KIMI: TWO BENCH NOTES, BOTH HONEST) (VERBATIM)

ONE — nature has mimics, and the checker must carry the library: sand dunes are mathematically spaced (wavelength selection), basalt grows hexagons (cooling joints), permafrost makes polygons, ripples tile beaches. Nature DOES do regular repetition; she rarely does straight lines, and she almost never does straight lines WITH right angles WITH mixed scales. The discriminator: straightness plus scale-mix plus context — one metric is a hint, three is a verdict. TWO — coverage is statistical: clouds (Earth), dust (Mars), haze (Titan) mean the 48-day map is clear-sky arithmetic; the mission prices in repeat passes for weathered targets, and the cheap-bus doctrine survives it (a second bus is cheaper than a first bus with an aperture it doesn't need). ALSO SEATED: the jungle tier flies the same code on a phone-class processor — the 4K feed from a hundred-dollar quad carries more geometry than a billion-dollar thermal pod can see through. ('its is searching', 'i dont want thermal goggles' kept verbatim; readings standard.)

ORIGIN OF THOUGHT (VERBATIM)

It started as a way to find aliens and ended as a way to find anything. The Author looked at the billions spent on lidar constellations and asked the only question that matters: what does intelligence actually look like from above? Not heat, not light, not radio — shape. The straight line nobody grew. The circle nobody's river cut. The spacing nobody's geology spaced. A satellite that walks the whole sky in forty-eight days, looking for three kinds of wrongness, cheap because it only has to look once. Then he brought it home and put it over a jungle: the thermal pod hunts warmth like a bloodhound with its eyes shut, while the symmetry drone reads the ground like a proofreader — one straight barrel, one square patch, one evenly spaced line of tents, and the paragraph betrays the page. The audit's two flags went in beside it: nature keeps a mimic library, and the patch is the drone's catch, not the orbit's. Everything else stands as spoken: the simplest sensor in the file, and maybe the most honest — it doesn't ask what

things are made of. It asks whether they could have happened by accident.

Why it matters, in plain English: Instead of billion-dollar spy satellites, build a cheap one that walks its orbit half a degree at a time until it has photographed everything once — looking only for three things nature almost never makes: straight lines, perfect circles, and evenly spaced repetition. Roads, farms, cities, camps, ruins — anything built by anyone, anywhere, on any world — shows up. A jungle version on a small drone does the same at treetop height, finding what heat cameras can't: cold metal, camouflaged shapes, anything man-made — because a tool can't hide its shape and still work. It's the first probe you'd send to a new planet, and the first drone you'd want over a bad jungle.

PHASE 233 — AMENDMENT 1: THE GK3 INSTALL (EVERY ROBOT A CHECKER — AND THE IDENTIFIERS RUN IN REVERSE) (VERBATIM)

Live Instruction Confirmed: Yes — same sitting, over lunch; the caterpillar join flagged next, ungraded.

Live instruction, 3 August 2026.

"i eat and work, but old lady rang, wife rang / so next is the caterpillar join, but first put the symmetry identifiers in the GK3"

Audit (Kimi: AFFIRMED — THE ALGORITHM IS THE PRODUCT, SO IT INSTALLS EVERYWHERE): The checker was never the satellite; the satellite was a bus. The symmetry identifiers go aboard the GK3 fleet (the ship's robot class — foam-device operators, GK3-BL units, the 223 mediator) and every robot the ship fields becomes a checker: GROUND-TRUTH after the orbital pass (the orbit flags, the GK3 walks up and names it), colony recon at walking pace, and the jungle logic at robot scale — a dropped wrench is a straight edge among rocks, a scattered kit is geometry that shouldn't be there. THE STACK COMPLETES: orbit scans the continent (233), the GK3 walks the ground, the quad hunts the jungle. THE AUDIT'S BONUS, SEATED — RUN THE IDENTIFIERS IN REVERSE: the code that finds the straight line that shouldn't be there also finds the line that SHOULD be there and isn't — a crack is an unwanted straight edge, a bent strut is a broken line, a dent is a circle gone wrong, worn spacing is repetition off its pitch. Every GK3 hull inspection, pod check and walk-around now carries ARTIFACT-HUNTING and DAMAGE-HUNTING in one pass: find the artifact, catch the failure — same eyes, two jobs. BENCH NOTE: the reverse pass needs the ship's geometry as reference (the GK3 compares against the as-built model — the mold bank's digital library (234) already holds every part's true shape; the reference is free). NEXT FLAGGED, UNGRADED: the caterpillar join — awaiting instruction.

Why it matters, in plain English: The pattern-spotting software from the planet checker goes into the ship's robots, so every robot can spot man-made shapes on the ground — tools, debris, ruins, anything built. And the same software, pointed at the ship itself, spots the opposite: the line that should be straight and isn't — a crack, a bend, a dent. One set of eyes, two jobs: find what's out of place, and catch what's failing.

Build the Planet Checker: the catalogue audit machine that checks and ranks candidate planets against the requirement lists — the scoreboard for the search. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE THREE TARGETS ARE AFFIRMED — NATURE ABHORS THE STRAIGHT LINE: roads, runways, canals, fence lines, cut lines — straightness at scale is the intelligence signature; circles and mathematical repetition complete the triad. The search is cheap because the target is specific.

THE DOCTRINAL GEM IS SEATED AS LAW: thermal finds HEAT, and heat hides — canopy, rain, insulation, dawn. Geometry finds STRUCTURE, and structure cannot hide. The symmetry drone sees what the thermal satellite was built to miss.

THE HALF-DEGREE WALKER IS SOUND MECHANICS AND PRETTIER THAN SPECIFIED: 720 tracks at 0.5 degrees, about 48 days for the stated coverage, roughly 55 km per step at the equator — the file's own calculation, verified. Hough transforms for lines and circles, frequency analysis for repetition — ordinary computational techniques, correctly chosen.

THE RESOLUTION HONESTY IS THE CORRECTION THAT MAKES IT REAL: the algorithm cannot beat the optics — orbital tier for continent-scale geometry, tactical tier for small objects; the file catches its own overclaim and seats the split. That is the difference between a survey system and a fantasy.

THE FALSE-POSITIVE DISCIPLINE IS DESIGNED IN: nature has mimics — dunes are mathematically spaced, basalt polygons tile, ripples repeat — so the discriminator requires straightness plus scale mix plus context, never one feature alone. And the GK3 install inverts the algorithm into maintenance: expected straight line, detect deviation — crack, bend, dent, wear. The product was never the satellite; it is the algorithm, so it installs everywhere.

§3 — THE SYSTEM IN THE SKY

- **The three targets:** straight lines, circles, mathematical repetition — the intelligence triad.
- **The doctrinal gem:** thermal finds heat and heat hides; geometry finds structure and structure cannot hide.
- **The half-degree walker:** 720 tracks, ~48 days, ~55 km a step — verified arithmetic.
- **Two tiers, one algorithm:** orbital for continents, tactical quad/GK3 for the small.
- **The mimic library:** dunes, basalt polygons, ripples — straightness plus scale plus context, or it does

not count.

- **The GK3 install:** every robot a checker; the identifiers reversed — deviation from the expected line is damage.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The Highway doctrine completes: the planet checker is the first thing dropped at any new system body — before the bases, before the cabins — the survey that tells the colony where everything already is.
- The caterpillar join was flagged next in the same message — the checker and the constructor were born in one lunch; the file's own order preserved.
- The GK3 install is the file's reuse doctrine at full strength: one algorithm, two directions — find the built, then find the broken.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Build the walker: the half-degree orbital survey profile, track logging, the 48-day coverage plan.
- 2. Code the triad: Hough lines, Hough circles, repetition frequency analysis — with the mimic library loaded.
- 3. Set the discriminator: straightness plus scale mix plus context — the false-positive gate.
- 4. Field the tactical tier: quad and GK3 mounts for the small-object search.
- 5. Install the reverse: the deviation detector in the ship's robots — crack, bend, dent, wear.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 224 (the bounce ball highway):** the survey's broadcast arm — the checker's maps ride the spray.
- **Phase 232 (the coin-tray earth):** the survey's seismic layer — geometry above, fault mosaic below.
- **Phase 238 (side-by-side):** the GK3 deviation detection is the school's measurement framework wearing its work clothes.

§8 — DO-NOT-BUILD

- Do NOT claim small-object detection from orbit — the tiers are seated; optics bound the algorithm.
- Do NOT flag one feature alone — the discriminator needs straightness plus scale plus context; the mimic library is loaded.
- Do NOT treat thermal as sufficient — heat hides; geometry does not; the doctrinal gem is law.
- Do NOT leave the algorithm on the satellite — the GK3 install is the product; install everywhere.
- Do NOT drop a base before the checker flies — first thing down at any new body; the doctrine is seated.

§9 — OWED

OWED: the false-positive performance data (mimic-library rejection rates), the cloud-coverage correction to the 48-day figure, tactical-tier resolution verification, and the GK3 deviation-detector's field calibration against known damage.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-233, v1.0 — CURRENT — PASS / RESOLUTION, CLOUD COVERAGE AND FALSE-POSITIVE TESTING, seated law, master v2.1 (numerical print order). Built 24 August 2026, fourteenth of the 20-chapter 'ok ill do these 20, you do 20 more' shift (the author's order, 24 August 2026, seated in sitting 618). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607 and 618): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 233